

Baacha Adithya

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Summary

Full Stack and AI systems engineer who independently owns production infrastructure end-to-end, at enterprise scale: 300–400x search latency reduction (8–12s → 20–30ms p99) across 1M+ documents, 80% cut in contract processing turnaround, 36% lift in NLP extraction accuracy — all shipped to production. Deep expertise in Java, Python, C++, distributed systems, GenAI/agent LLM pipelines, Elasticsearch, Kafka, and Kubernetes; track record of authoring design docs and mentoring engineers.

Experience

Oracle – India

June 2024 – Present

Application Solution Developer | Java, Spring Boot, Elasticsearch, Python, GenAI, Kafka

- Architected Elasticsearch-based **full-text search** replacing Oracle SQL LIKE queries on 1M+ contract records, reducing search **p99 latency from ~8–12s to ~20–30ms** (300–400x improvement), enabling near-real-time retrieval for contract and deal teams.
- Designed and owned a **Kafka-based CDC data pipeline** synchronizing 5,000+ contract records daily from Oracle DB to Elasticsearch with sub-minute indexing latency, ensuring **high availability** and data consistency across distributed search and reporting services.
- Built **agentic LLM workflows** (LangChain, RAG, OpenAI APIs) for automated clause insertion and contract intelligence, and developed a GenAI-based NLP key term extractor achieving **36% accuracy improvement** over rule-based methods; authored technical design documents and contributed to cross-team code reviews.
- Developed **vector search engine** for legal clause prediction— reducing contract drafting turnaround **80%** (from 1–2 weeks to 1–2 days) across 1M+ contract documents.
- Implemented **MLOps CI/CD pipelines** for continuous model training, deployment, and monitoring; maintained production **SLA reliability** with on-call support. Collaborated cross-functionally with platform and shared infrastructure teams, led code reviews.
- Designed complex **filtered query logic** (order number, status flags, Boolean conditions) over large-scale contract data, and built **Spring Boot microservices** for reporting, filtered export, and REST-based UI integration — enabling business teams to pinpoint exact record subsets across millions of records without manual filtering.
- Implemented **Redux-based state management** to cache and retrieve related-order data across the UI, eliminating redundant API calls and keeping order state consistent across components during navigation.
- Built **dynamic Oracle JET UI components** with server-side pagination to handle large result sets efficiently, cutting time-to-find for business users navigating high-volume search results and directly improving team productivity.
- Designed **fault-tolerant, scalable** Spring Boot microservices with JPA/Hibernate for reporting and REST-based UI integration; deployed via **Helm/Kubernetes** with service gateways, config isolation, and CI/CD automation for production-grade rollouts.

Meta XP Private Limited – Remote [Live Demo](#)

Oct 2022 – Nov 2022

Data Science Intern | Speech Recognition, NLP, Python

- Built an end-to-end ML pipeline using Google Speech-to-Text (ASR) to transcribe and analyze video audio at scale; performed structured **EDA on speaking patterns** (filler words, pitch, speech rate) with measurable output evaluation metrics for the public speaking assessment product.

Indian Institute of Science (IISc) Bangalore – Research Intern

Jul 2022 – Aug 2022

- Applied Python-based computational modeling using LAMMPS to simulate Lennard-Jones fluid dynamics; analyzed vapor-liquid coexistence and equilibrium pressure/density through structured data pipelines.

Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL

Backend & Systems: Spring Boot, Microservices, REST APIs, Distributed Systems, Fault Tolerance, Scalability, JPA/Hibernate

AI / GenAI / ML: Generative AI, OpenAI APIs, LangChain, RAG, Vector Search, Agentic Workflows, MCP, PyTorch, TensorFlow, scikit-learn, XGBoost, MLOps, FastAPI, Pydantic

Data & Cloud: Elasticsearch, Kafka, PostgreSQL, Redis, Oracle SQL, OCI, GCP, AWS, Azure, Docker, Kubernetes, Helm

DevOps & Practices: CI/CD, Jenkins, GitHub Actions, ArgoCD, Git, Code Reviews, Design Documents (RFCs), On-Call/SRE

Frontend: React, Oracle JET, TypeScript, JavaScript, HTML, CSS

Core Strengths: System design, cross-team collaboration, production ownership, mentorship, rapid adaptability

Education

National Institute of Technology Calicut (NIT Calicut) | B.Tech, Chemical Engineering | CGPA: **9.2** | May 2024

Projects

Image Captioning — Vision-Language System (2022): Architected an end-to-end PyTorch pipeline coupling a ResNet-18 spatial encoder with a custom 4-layer, 16-head Transformer decoder for sequential caption generation over an 8K+ token vocabulary, using masked Cross-Entropy loss for padded-sequence handling and Top-K sampling inference for fluent, well-terminated captions.

Certifications

Deep Learning Specialization, Andrew Ng – 2023 | Oracle Data Science | Oracle Vector Search | Fusion Agent Studio – 2025

Honors and Awards

- Gold Medalist** – Final Year Branch Topper: Highest CGPA in B.Tech Chemical Engineering, NIT Calicut.
- Rockstar Gold Award, Oracle (Q1 2025):** Highest-tier organizational award for exceptional performance and significant contributions to OCIP platform enhancement.
- Competitive Programming:** LeetCode contest rating 1726.

Interests: Chess, Cricket, Cooking, Travelling, Stock Analysis